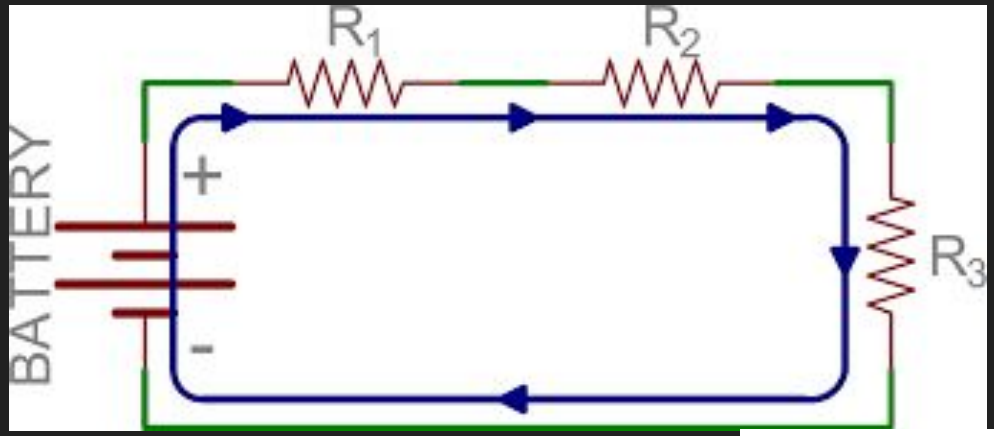
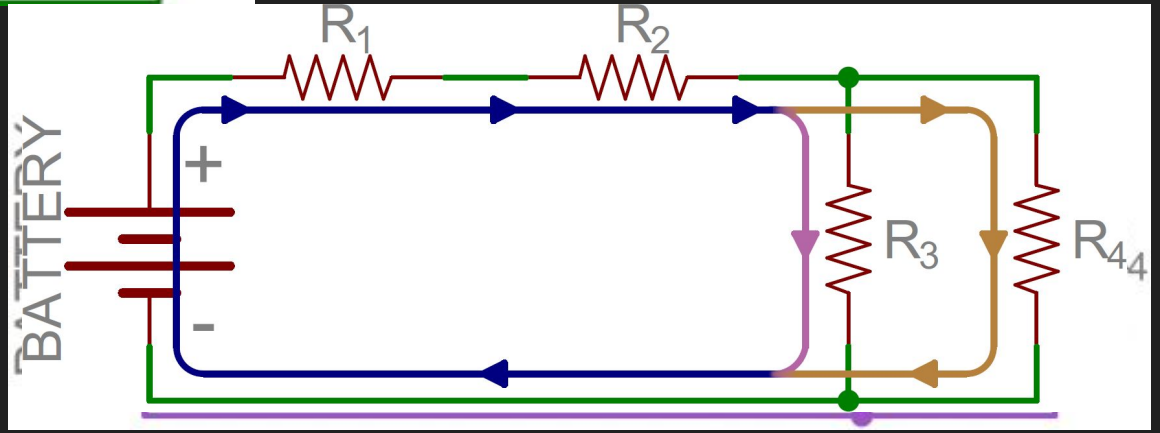
Digital Electronics Interfacing

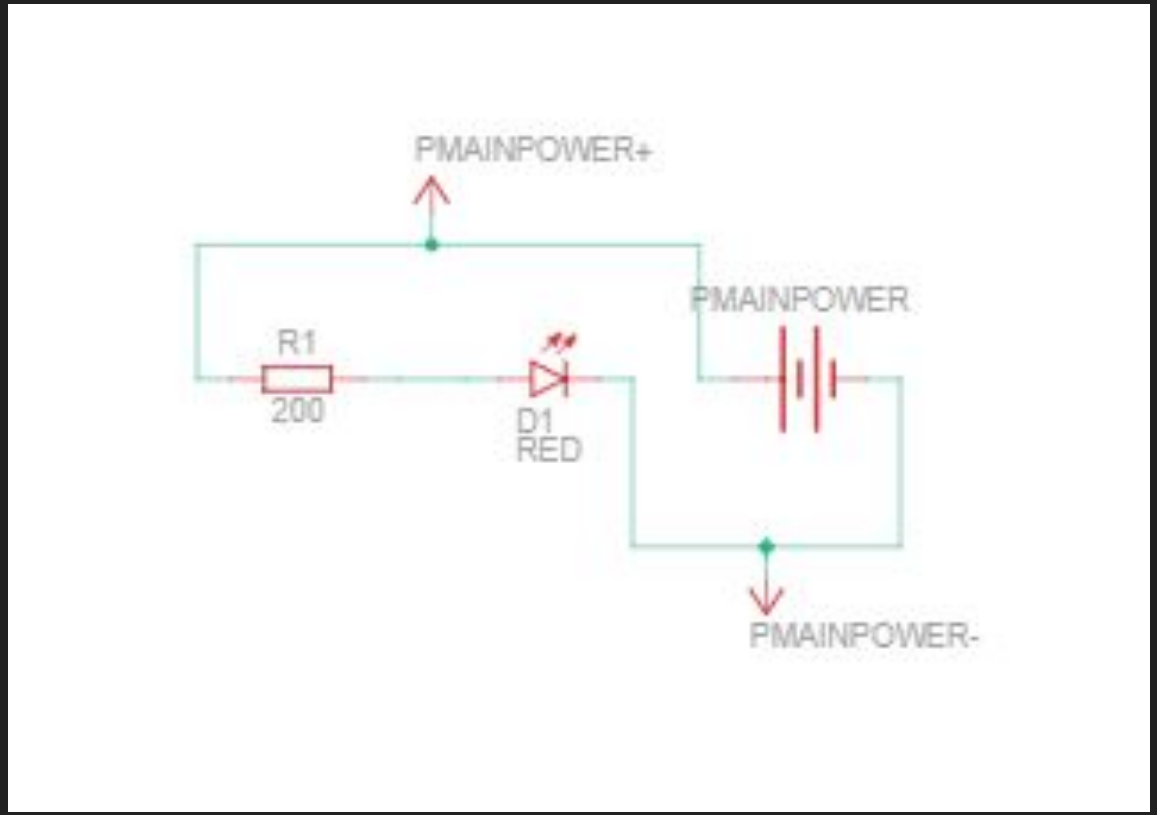
Lesson 4

Series and Parallel Circuits

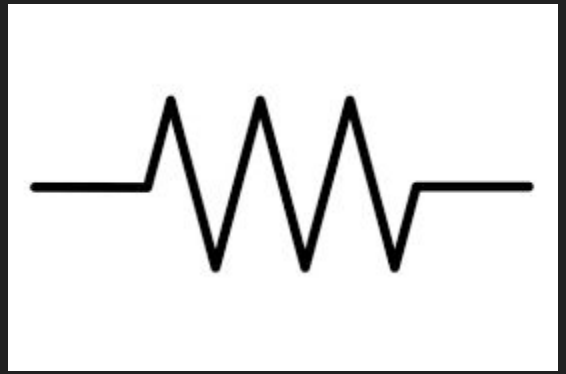


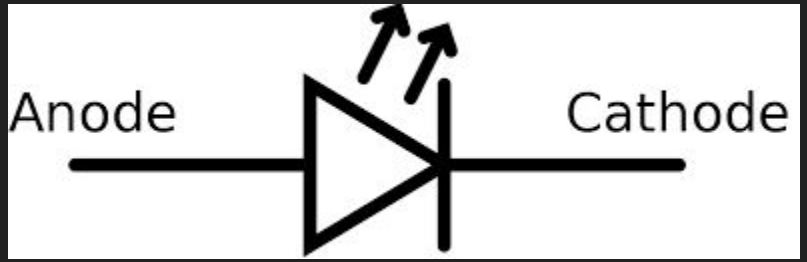


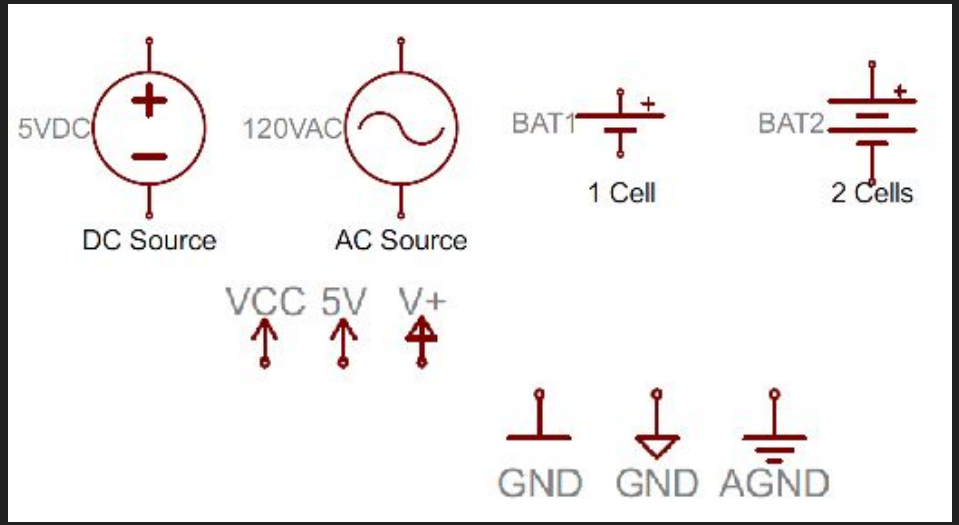
Schematic Series Circuit



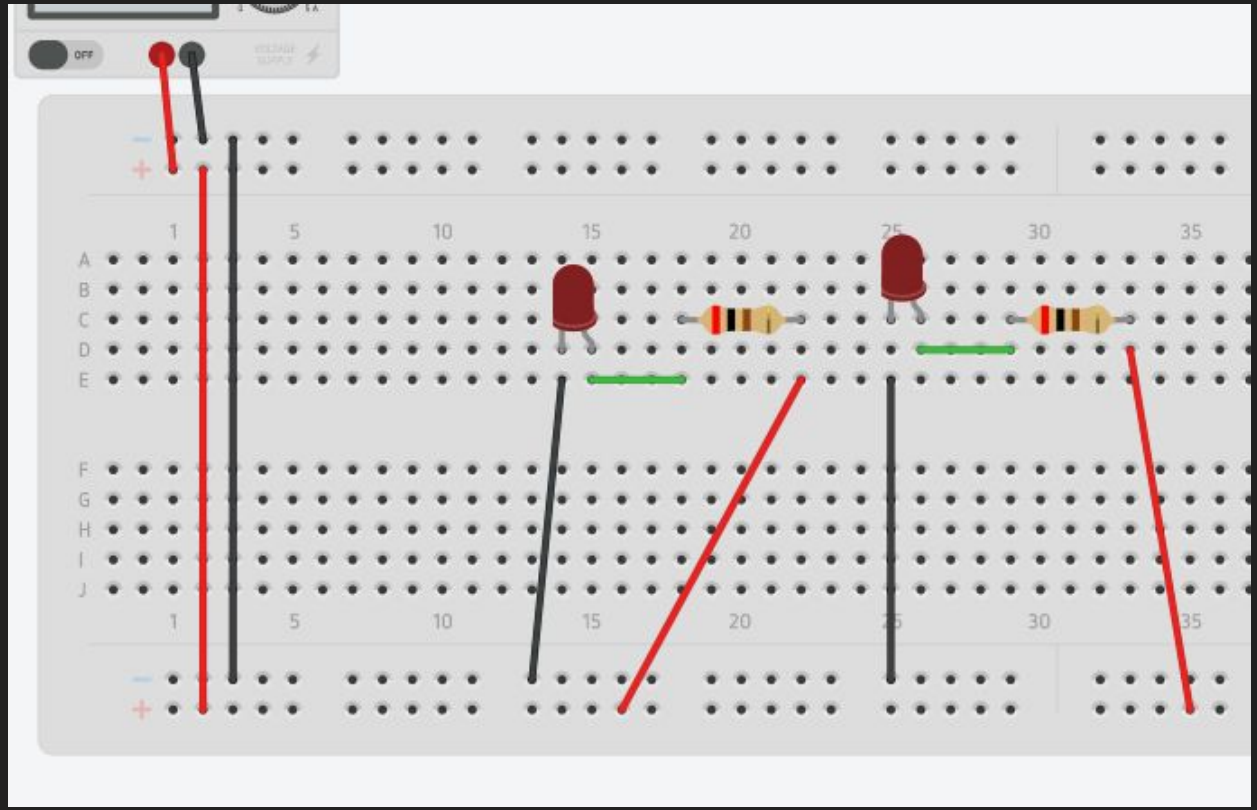
Schematic Symbols



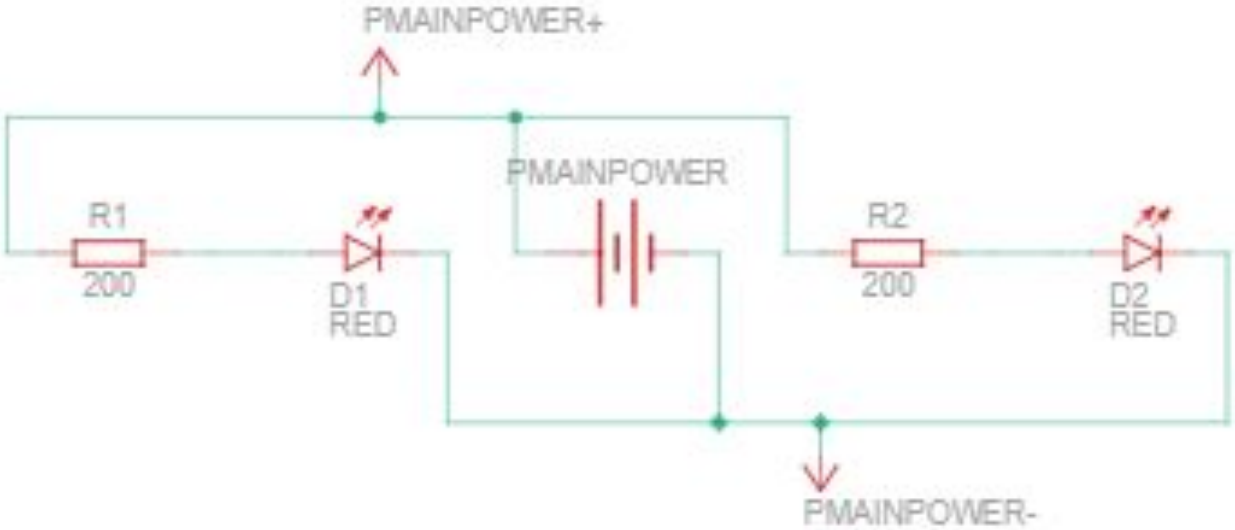




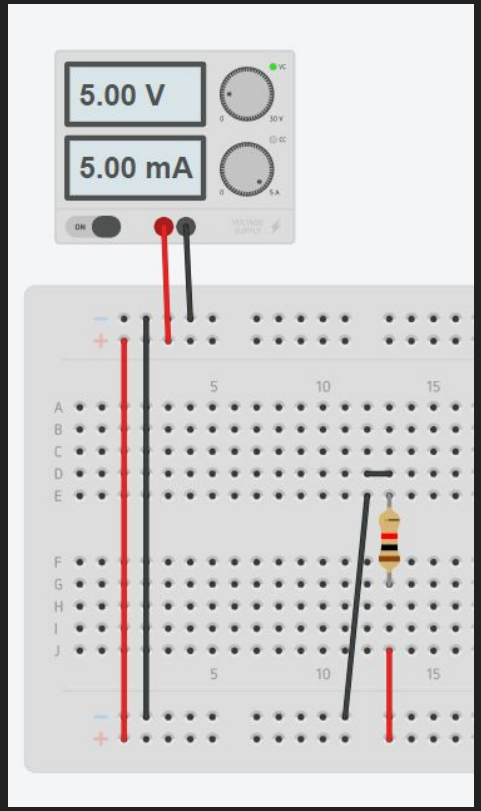
Parallel Circuit in Tinkercad



Parallel Circuit Schematic in Tinkercad



Current in Series and Parallel Circuits



Resistors are 1 Kohm

Series Circuit Resistance=R1+R2+R3+...

Total resistance=sum of all the resistances in the series circuit

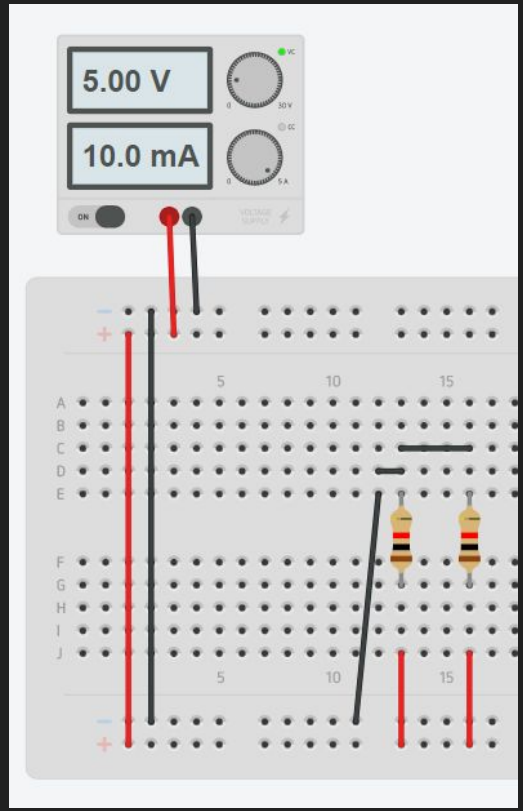
Rt=1 Kohm=1000 ohms

I=V/R=5V/1000=0.005A=5mA

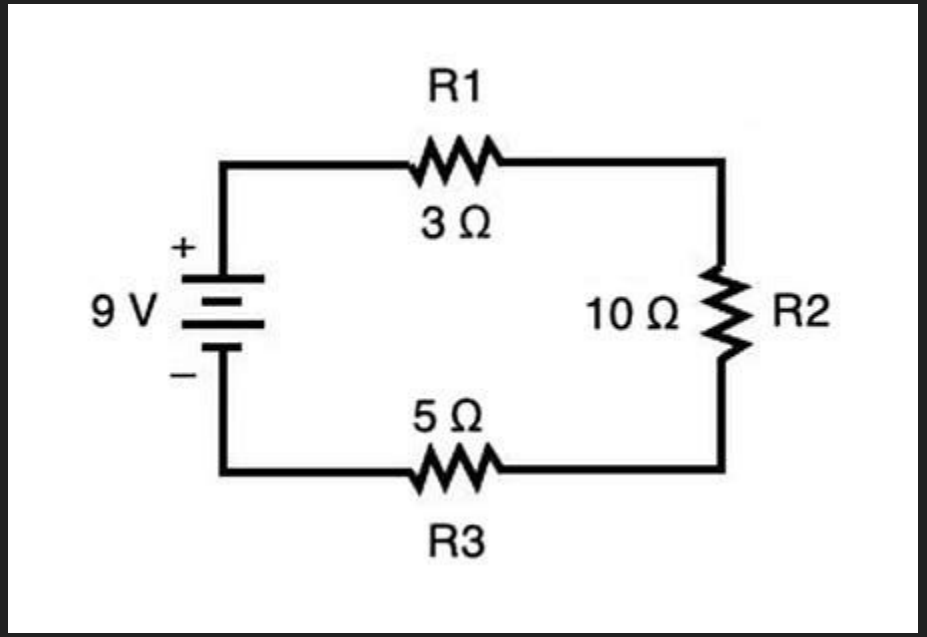
1/Rt=1/R1+1/R2+....

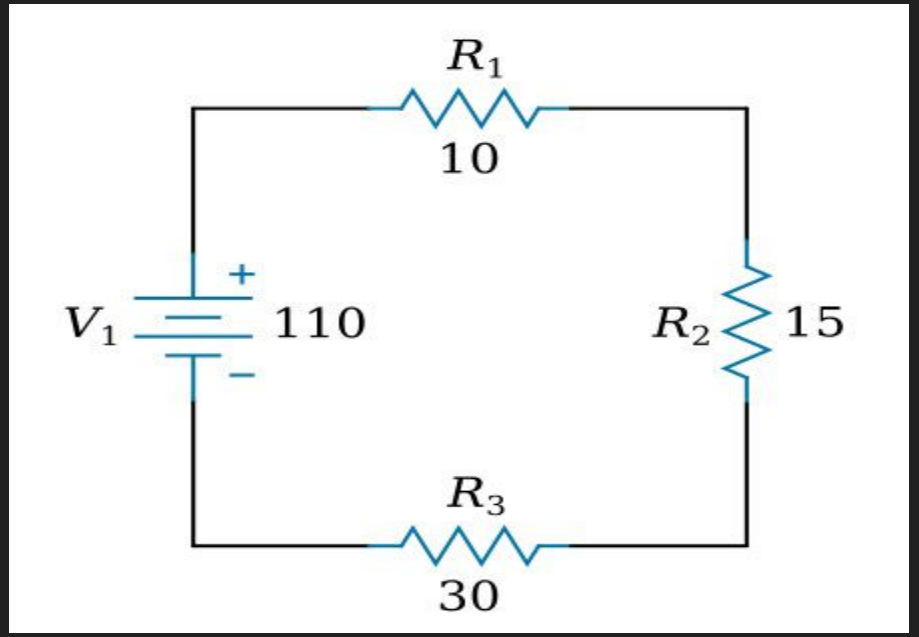
1/Rt=1/1+1/1
1/Rt=1+1
1/Rt=2
Rt=½
Rt=0.5 Kohm=500 Ohms

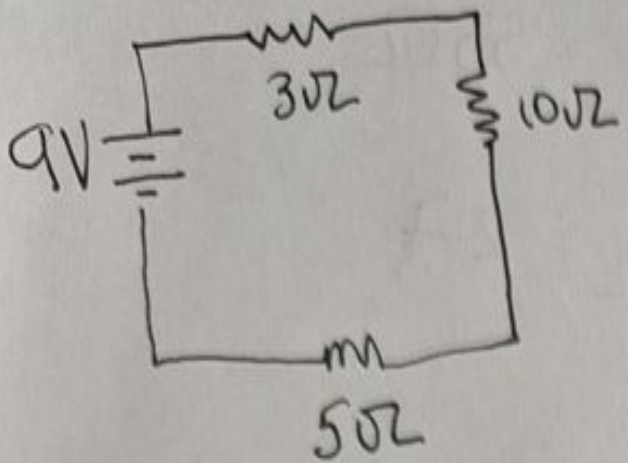
I=V/R=5V/500Ohms=.01A=10mA



Examples-Series Resistance





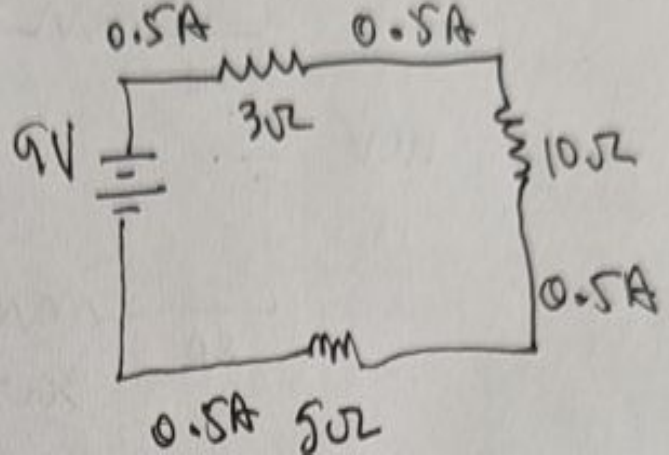


$R_T = 3 + 10 + 5 = 18\Omega$

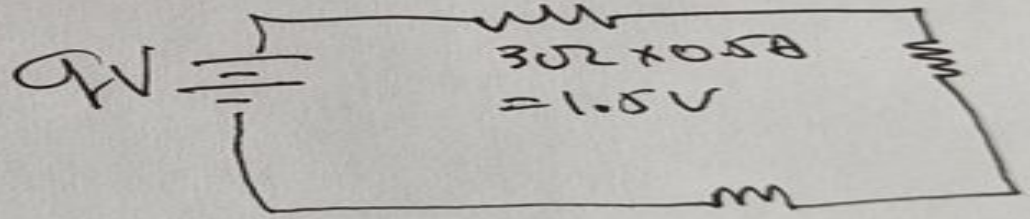
$I = \frac{V}{R} = \frac{9V}{18\Omega} = 0.5A$

Current is the same everywhere in a series circuit

$\therefore$

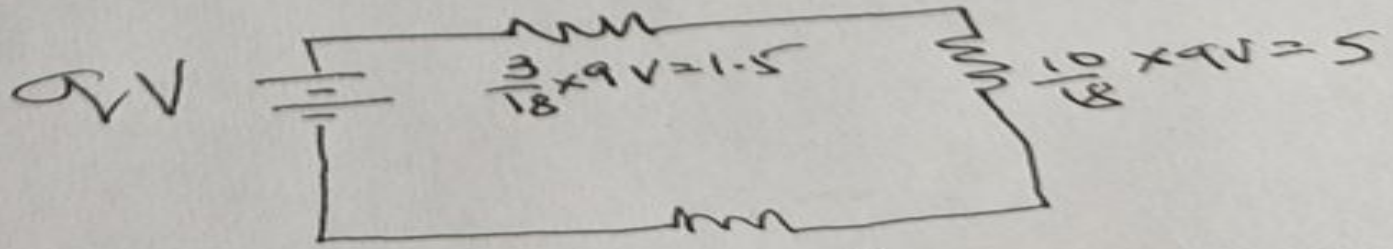


Voltage Drops → proportional to resistance relative to ~~total~~ resistance
→ must equal total voltage

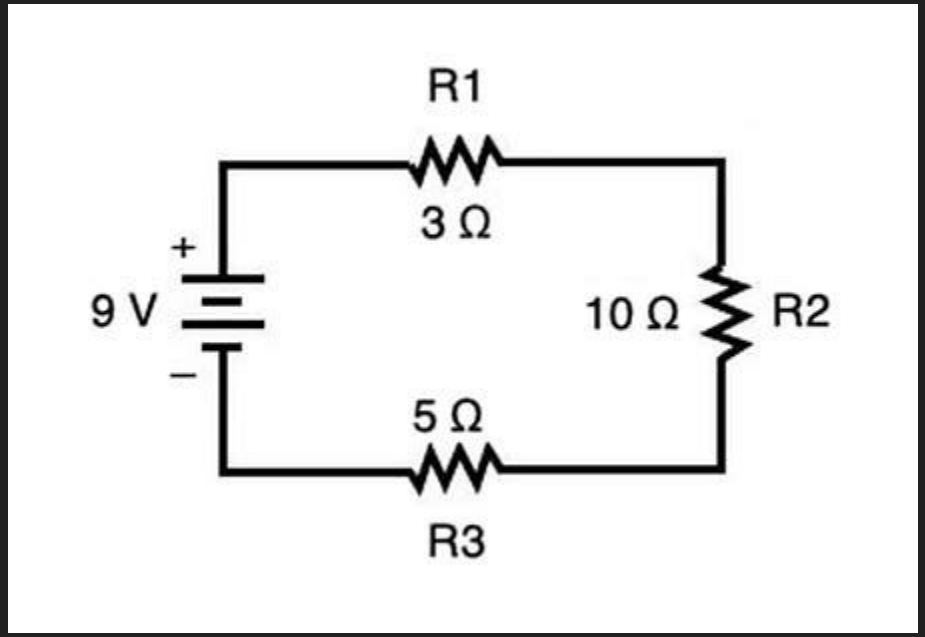


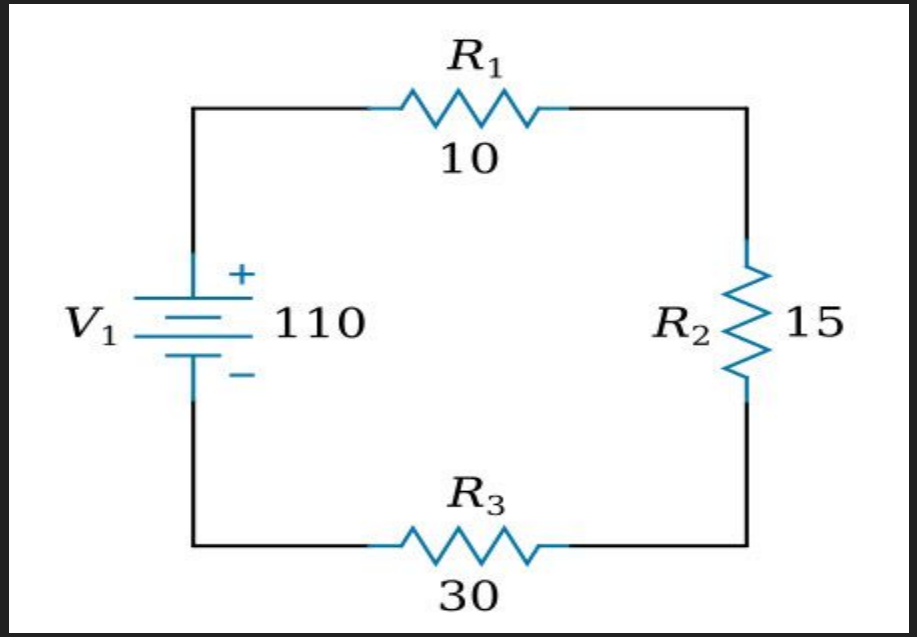
$V_T = 1.5 + 5 + 2.5 = 9V$

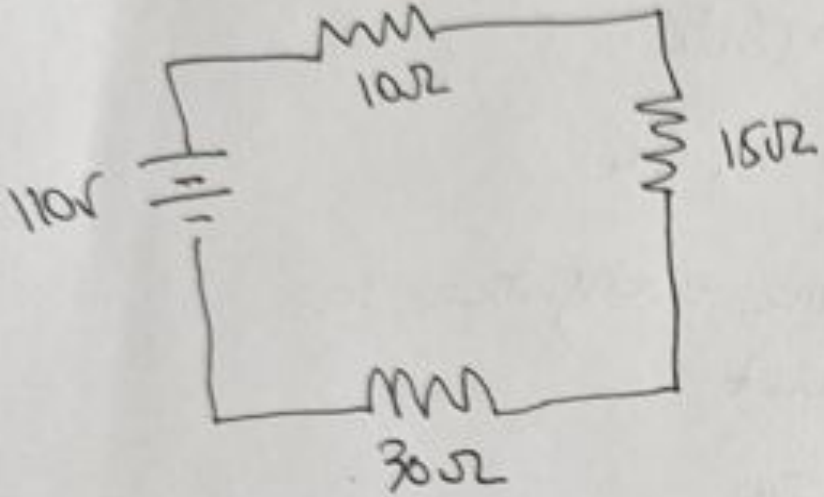
or



Examples-Series Current



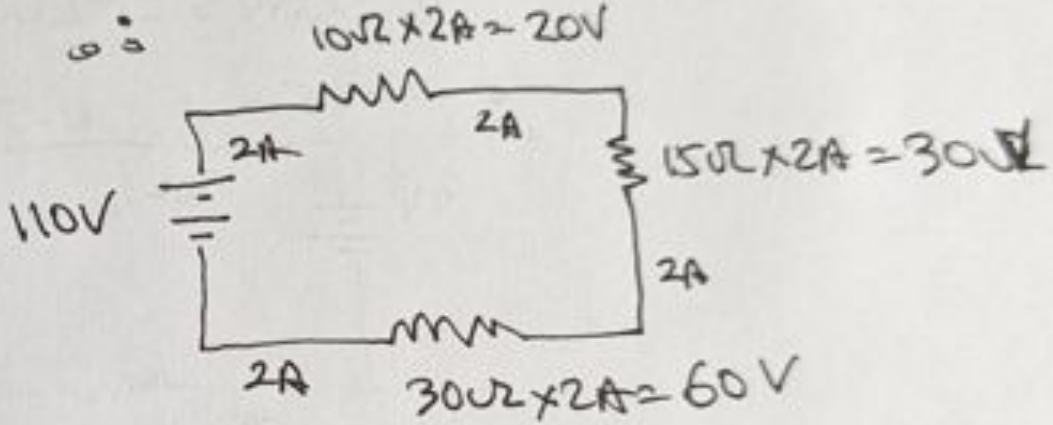




$R_T = 10 + 15 + 30 = 55\Omega$

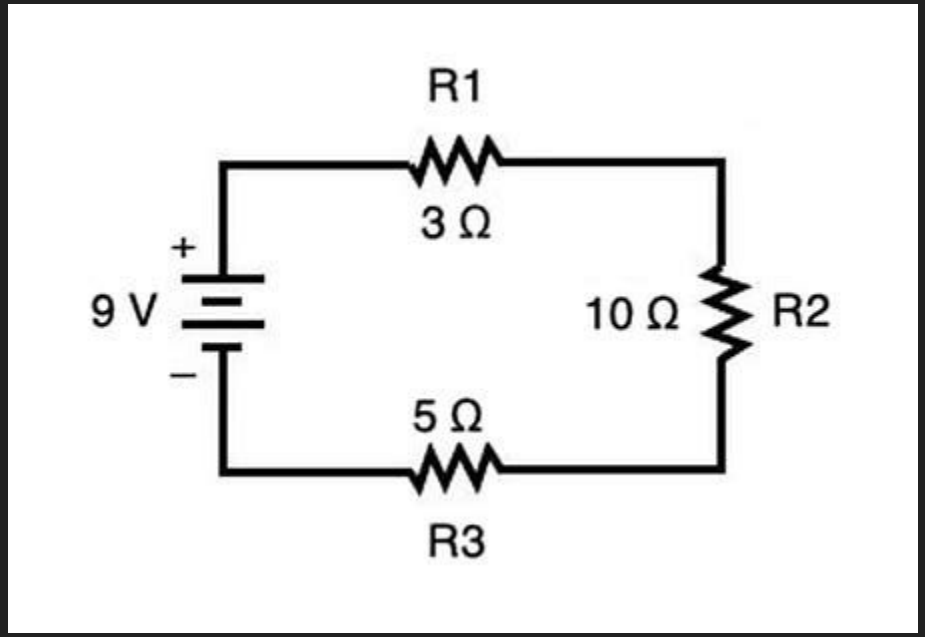
$I = \frac{V}{R} = \frac{110V}{55\Omega} = 2A$

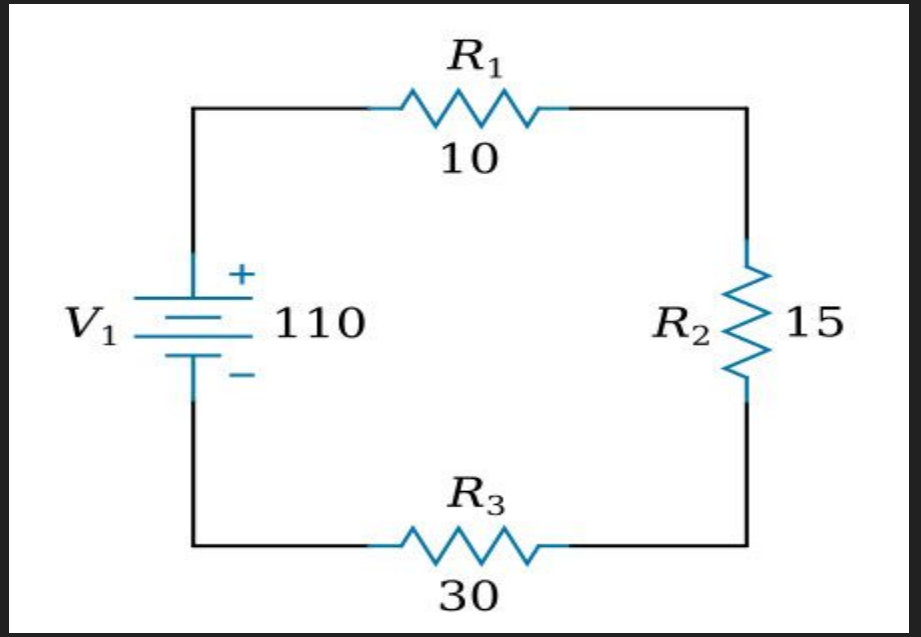
∴



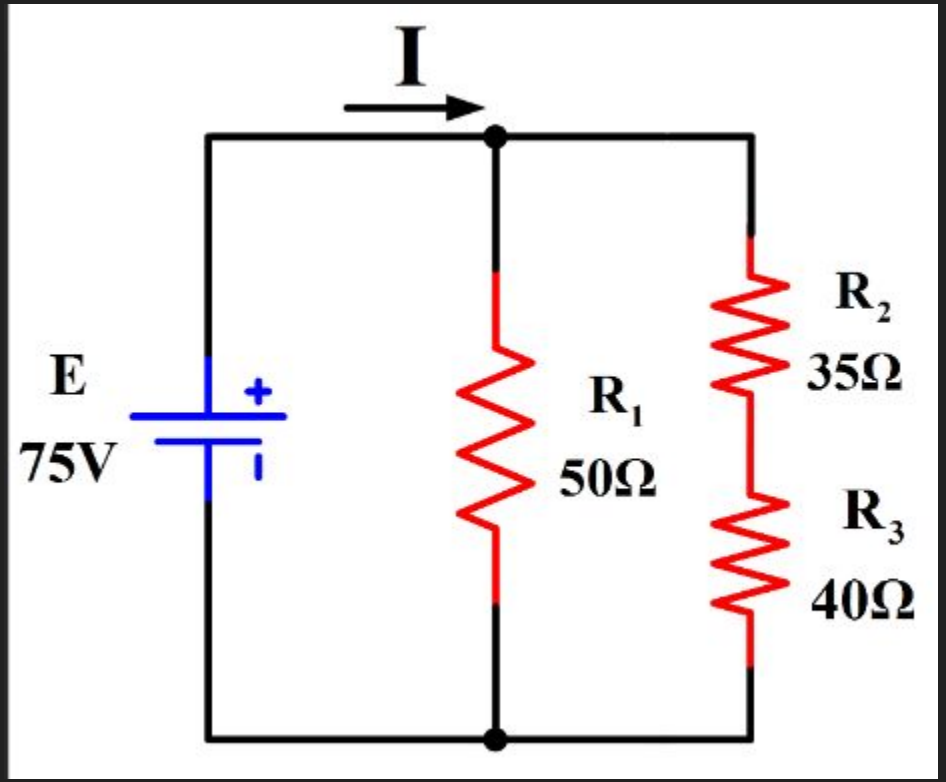
$V_T = 20V + 30V + 60V = 110V$

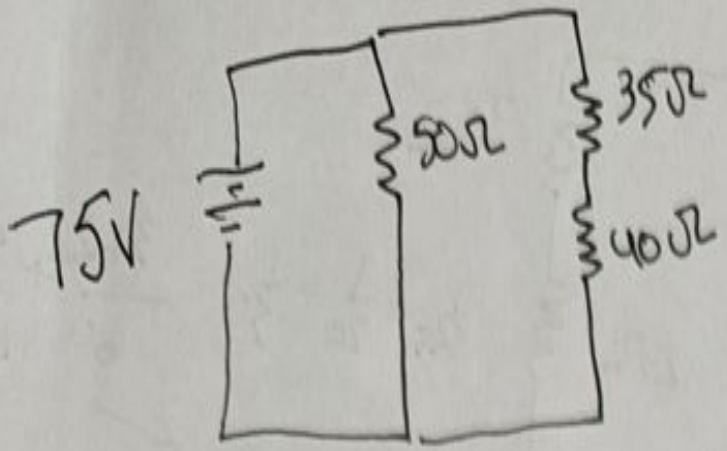
Examples-Voltage Drops

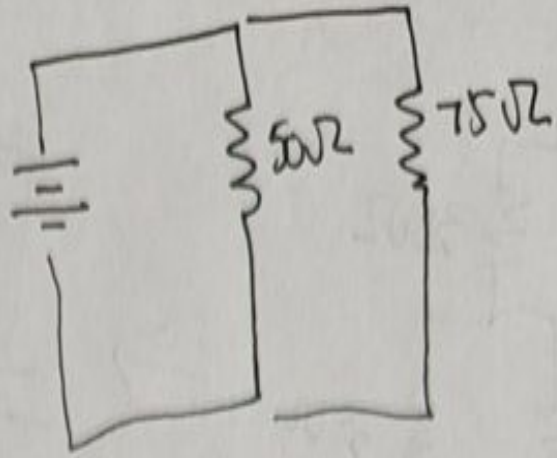




Parallel Circuit-Resistance







$$\frac{1}{R_T} = \frac{1}{50} + \frac{1}{75}$$

$$\frac{1}{R_T} = \frac{3+2}{150}$$

$$\frac{1}{R_T} = \frac{5}{150}$$

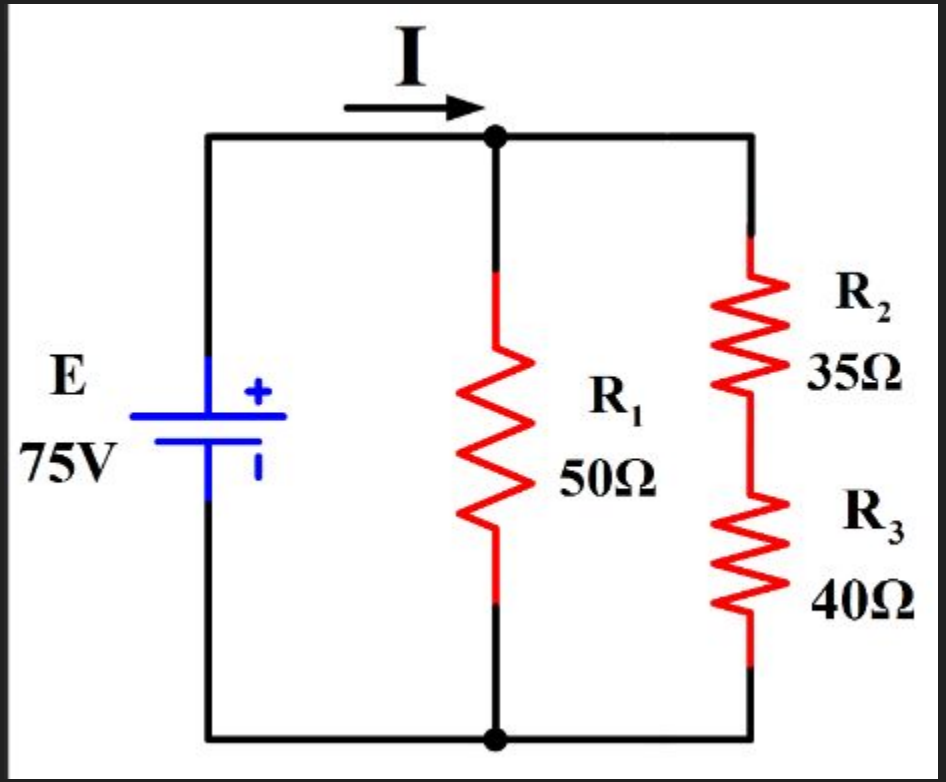
$$R_T = \frac{150}{5} = 30\Omega$$

$\therefore V = IR$

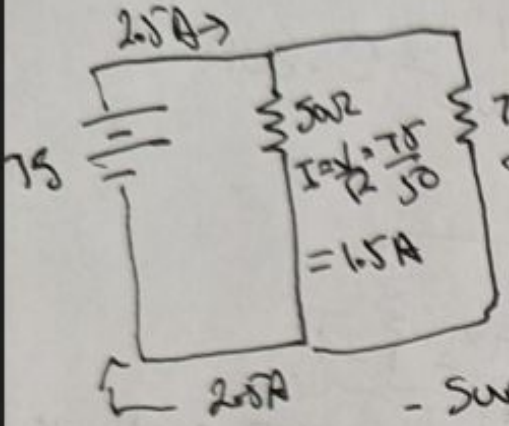
$$I = \frac{V}{R} = \frac{75V}{30\Omega} = 2.5A$$

2.5A →

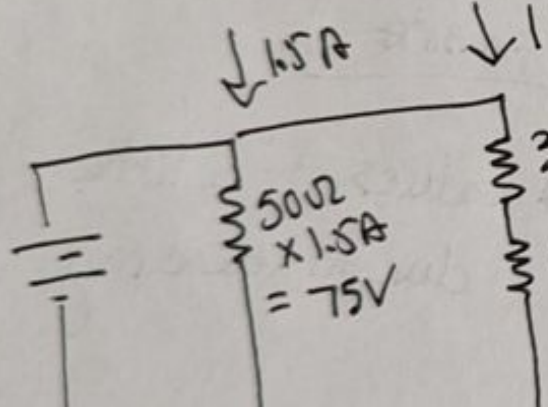
Parallel Circuit-Current



$I = \frac{V}{R} = \frac{75V}{30\Omega} = 2.5A$

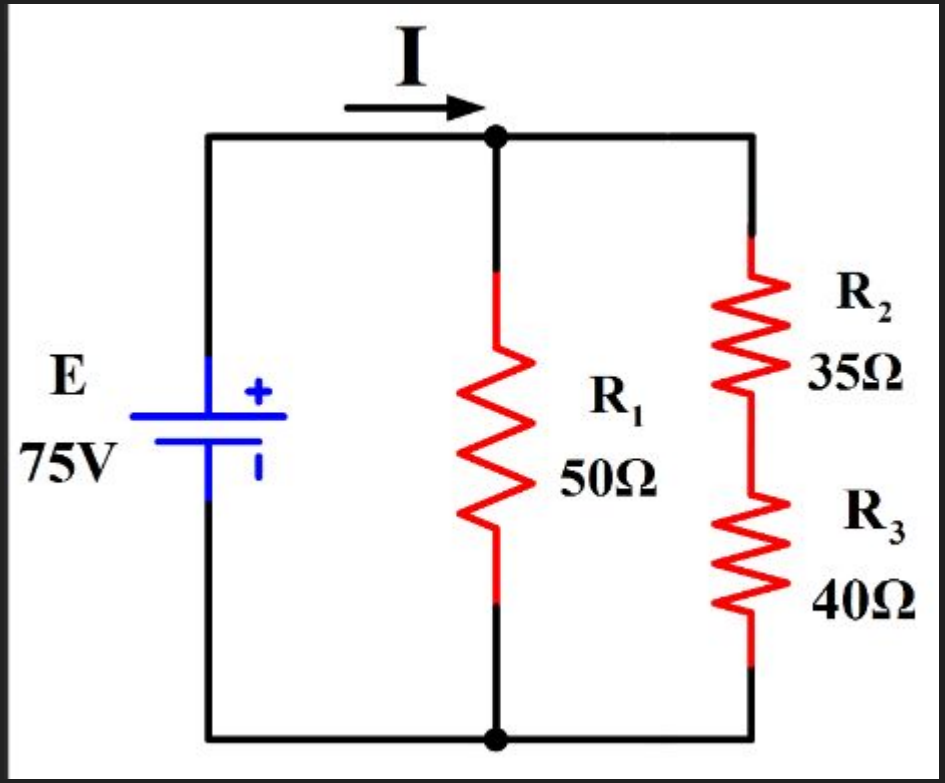


- sum of current in each branch equal to total current in the circuit

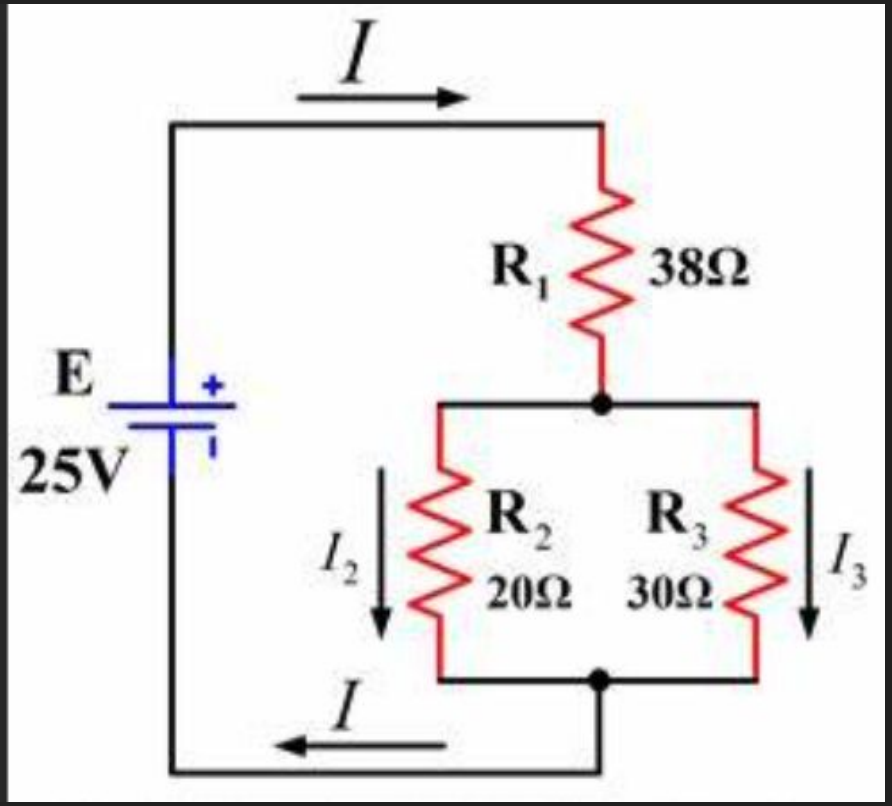


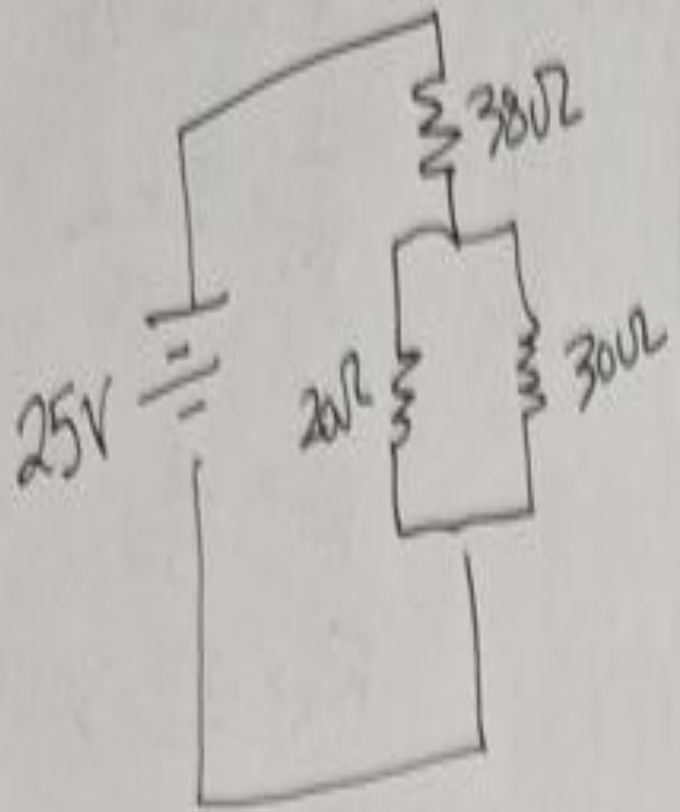
35Ω × 1A = 35V
40Ω × 1A = 40V
} 75V → sum of voltage drops equals voltage into series circuit branch

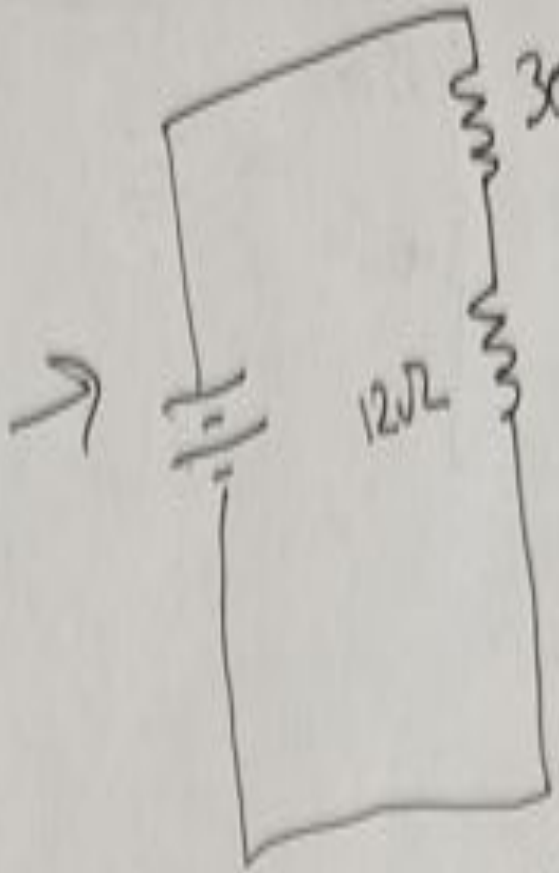
Parallel Circuit-Voltage Drops



Parallel Circuit-Resistance





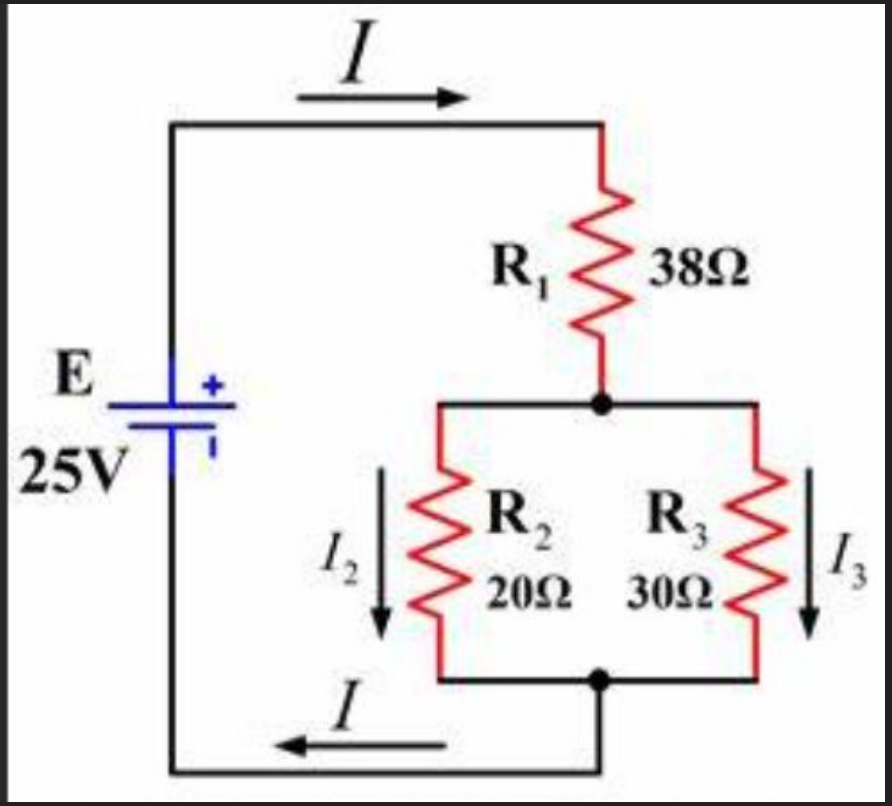


$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{30} = \frac{3+2}{60} = \frac{5}{60}$

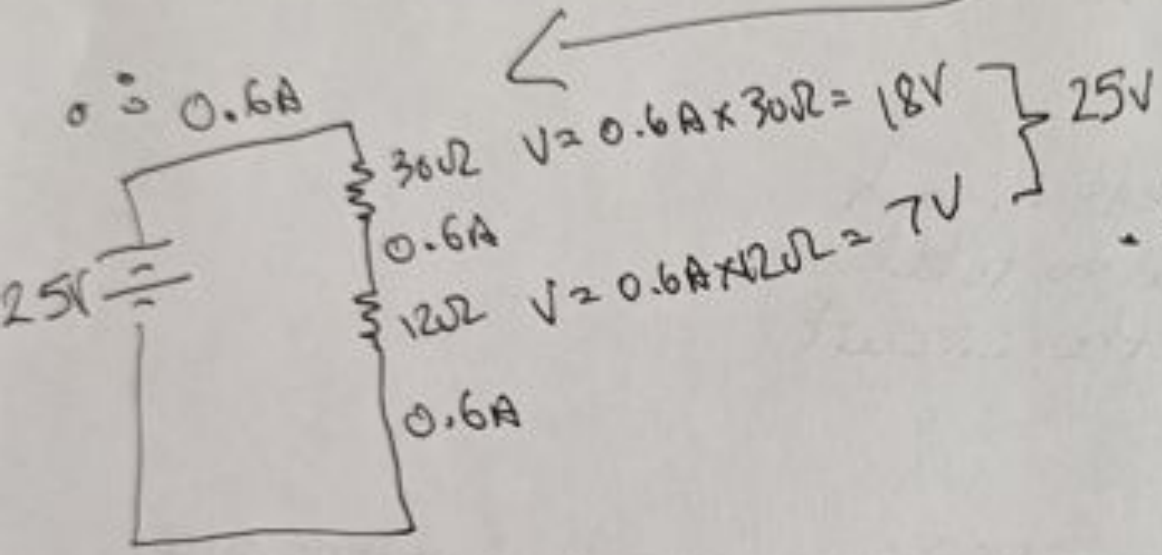
$R_T = 12\Omega$

$R_T = 30\Omega + 12\Omega = 42\Omega$

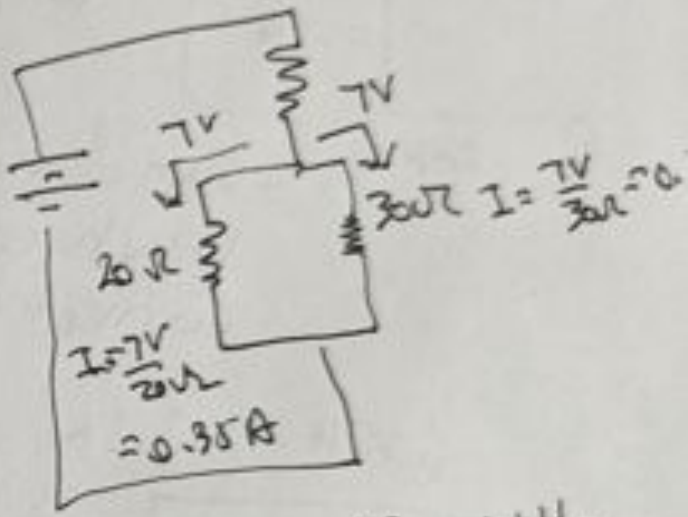
Parallel Circuit-Current



$\therefore I = \frac{V}{R} = \frac{25V}{42\Omega} = 0.60A$

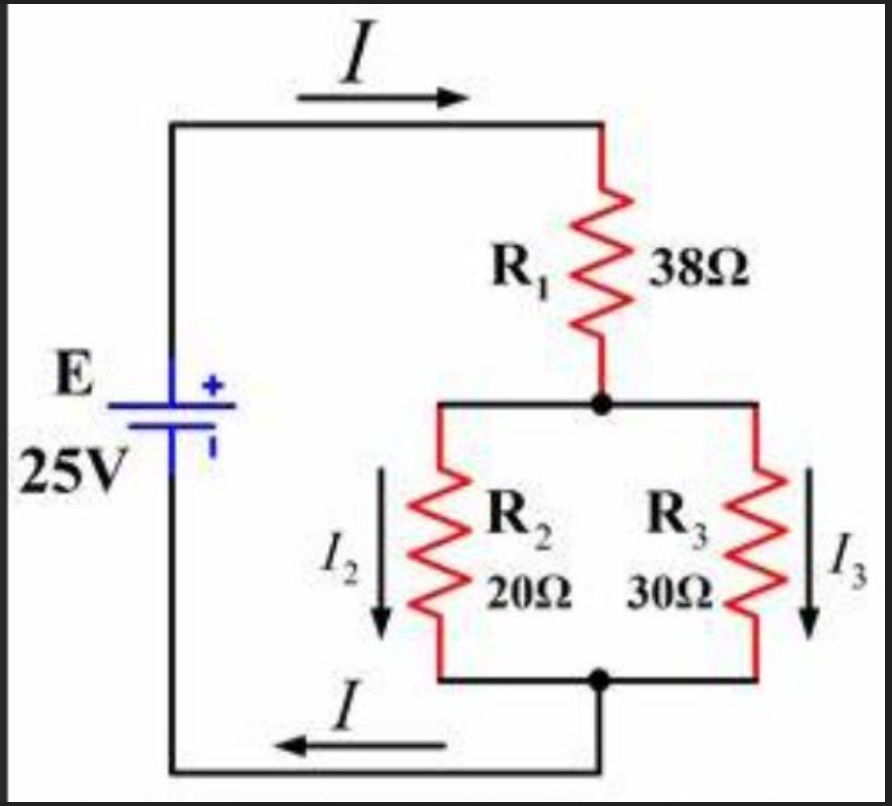


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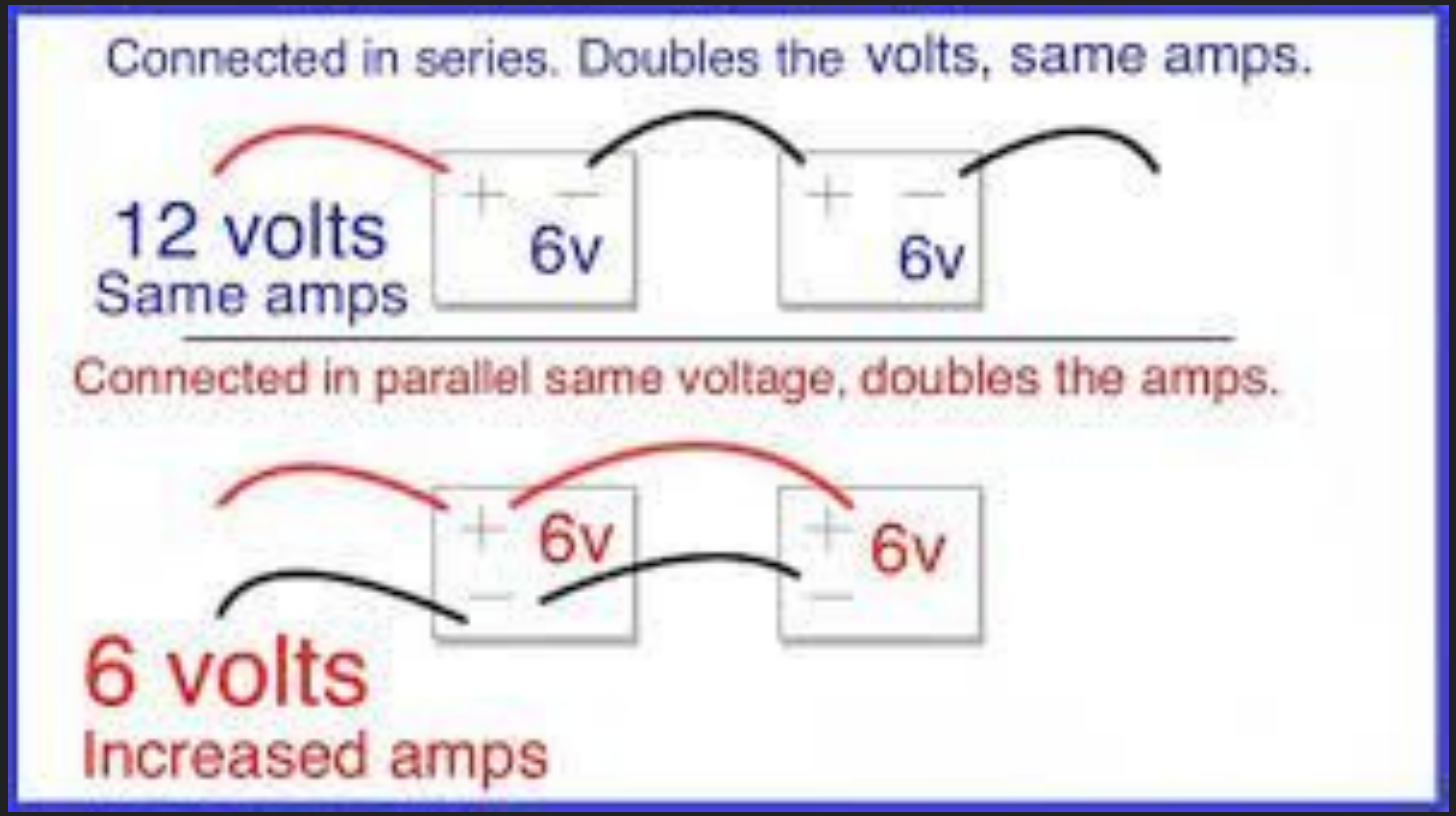


* values off a little due to rounding

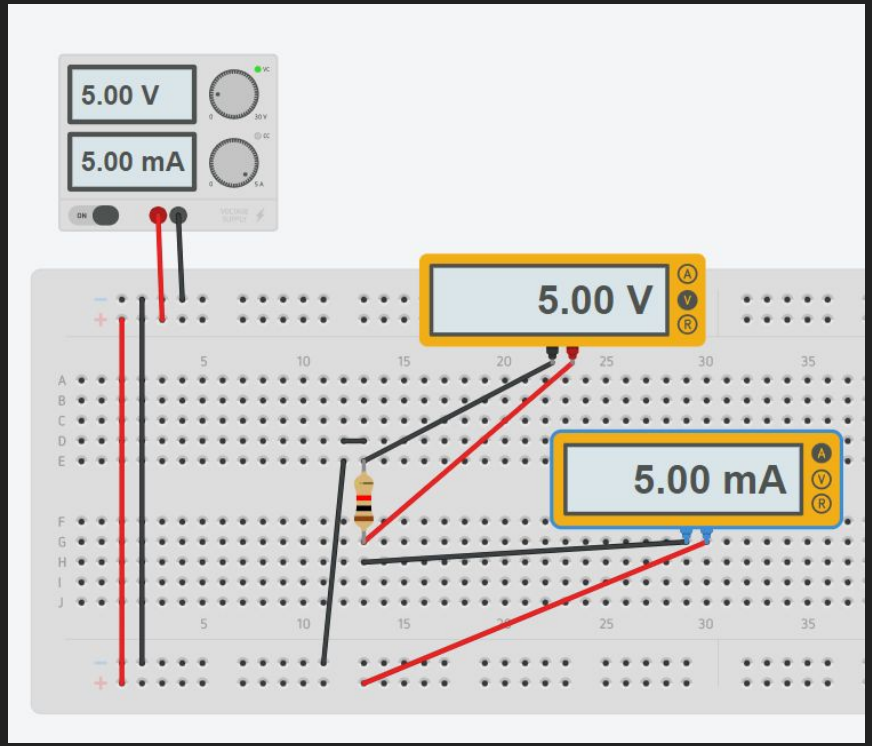
Parallel Circuit-Voltage Drops



Voltages in Series and Parallel



Measuring Current and Voltage Using a Multimeter in Tinkercad



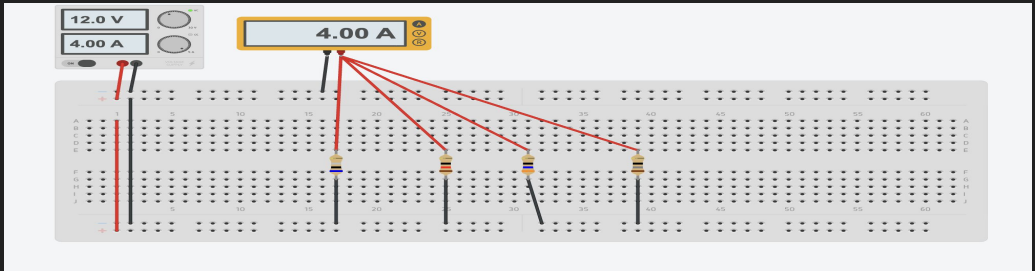
Questions and Exercises

1. Complete the worksheet (questions 1,2,3,4,5,6,7,8,9,10,11,12,13,14ab,15bc,16,18,19,20)

   https://docs.google.com/document/d/1rqThFTgd19KWBIMlVz-eJ5nxkPUFg3Ab/edit?usp=sharing&ouid=109368619532564711558&rtpof=true&sd=true

   https://docs.google.com/document/d/1ghqjteEpwlji8Z4F9rWXz-hb2qunRNKL/edit?usp=sharing&ouid=103567386690681902633&rtpof=true&sd=true

2. Create the circuit in question 7 of the assignment worksheet in Tinkercad. Insert the Multimeter in the same place to measure current. Embed a screenshot of your circuit with the results below.



Solutions

#1) a) $R_T = 7 + 5 + 2 = 14\Omega$

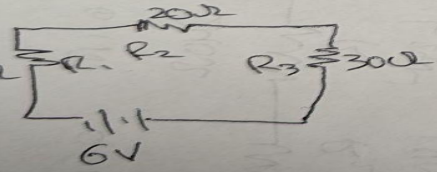
b) $R_T = 2 + 5 = 7\Omega$

c) $R_T =$ 1/R=½+1/7+⅕→R=70/59=1.19 Ohms

#2) a) $V_T = 3 + 4 + 6 = 13V$

b) $V_T = 6 + 6 = 12V$

#3)



$V = IR$

$6 = I((10+20+30)$

$6 = 60I$

$I = \frac{6}{60} = 0.1A$

$\therefore I_1 = I_2 = I_3 = 0.1A$

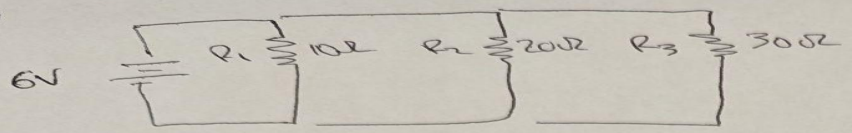
$\therefore V_1 = (0.1)(10) = 1V$

$V_2 = (0.1)(20) = 2V$

$V_3 = (0.1)(30) = 3V$ +

$6V$

#4)



$\frac{1}{R_T} = \frac{1}{10} + \frac{1}{20} + \frac{1}{30} = \frac{6+3+2}{60} = \frac{11}{60}$

$R_T = \frac{60}{11} = 5.45\Omega$

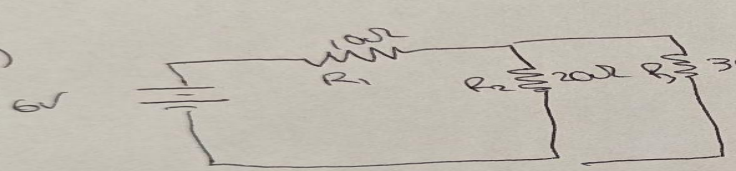
$\therefore I = \frac{V}{R} = \frac{6V}{5.45\Omega} = 1.1A$

$I_1 = \frac{6V}{10\Omega} = 0.6A \quad I_2 = \frac{6V}{20\Omega} = 0.3A \quad I_3 = \frac{6V}{30\Omega} = 0.2A$

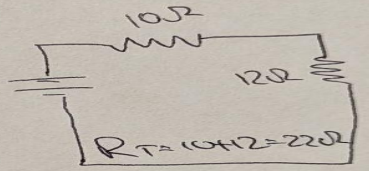
$I_{total} = 0.6 + 0.3 + 0.2 = 1.1A$

$V_1 = 6V \quad V_2 = 6V \quad V_3 = 6V$

#5)



→



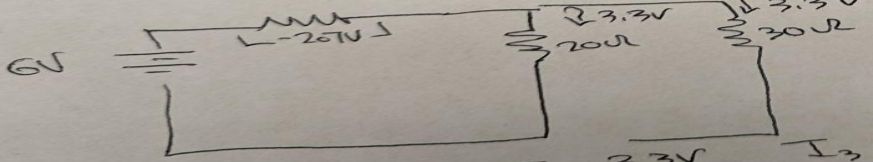
$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{30} = \frac{3+2}{60} = \frac{5}{60}$

$R_T = \frac{60}{5} = 12\Omega$

$I = \frac{V}{R} = \frac{6V}{22\Omega} = 0.27A$

$V_1 = 0.27A \times 10 = 2.7V$

$6V - 2.7V = 3.3V$



$I_1 = 0.27A$

$I_2 = \frac{3.3V}{20\Omega} = 0.165A$

$I_3 = \frac{3.3V}{30\Omega} = 0.11A$

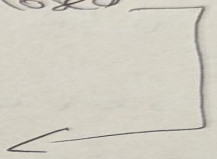
$I_T = 0.165 + 11 = 0.275A$

#6) $\frac{1}{R_T} = \frac{1}{6} + \frac{1}{12} + \frac{1}{36} + \frac{1}{18} = \frac{6+3+1+2}{36} = \frac{12}{36}$

$R_T = \frac{36}{12} = 3\Omega$

#7) $I = \frac{V}{R} = \frac{12V}{3\Omega} = 4A$

#8)
a) short circuit when switch closed
b) " "
c) proper working circuit
d) " "



#9) $V = IR$

$120V = (0.5A)R$

$R = \frac{120}{0.5} = 240\Omega$

$50 + X = 240$

$X = 240 - 50 = 190\Omega$

#10)

#11) $V = IR$

$12V = (5A)R$

$R = \frac{12}{5} = 2.4\Omega$

#12) $I = \frac{V}{R} = \frac{12V}{3\Omega} = 4A$

#13) $\frac{1}{R_T} = \frac{1}{3} + \frac{1}{R}$

$\frac{1}{2.4} = \frac{1}{3} + \frac{1}{R}$

$\frac{1}{R} = \frac{1}{2.4} - \frac{1}{3} = \frac{3 - 2.4}{7.2} = \frac{0.6}{7.2}$

$R = \frac{7.2}{0.6} = 12\Omega$

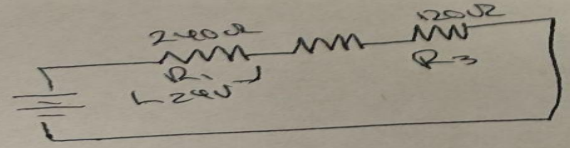
#14) a) $R_T = 125\Omega + 225\Omega = 350\Omega$

b) $I = \frac{V}{R} = \frac{50V}{350\Omega} = 0.14A$

#15) b) $R_T = 15\Omega + 21\Omega + 24\Omega = 60\Omega$

c) $I = \frac{V}{R} = \frac{12V}{60\Omega} = 0.2A$

#16)



a) $I_1 = \frac{V}{R} = \frac{24V}{240\Omega} = 0.1A$

b) $R = \frac{V}{I} = \frac{40V}{0.1A} = 400\Omega$

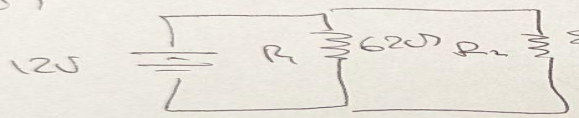
c) $R_2 = 400 - 240 - 120 = 40\Omega$

#18) $\frac{1}{R_T} = \frac{1}{12} + \frac{1}{18}$ $\frac{1}{R_T} = \frac{3+2}{36} = \frac{5}{36}$

$R_T = \frac{36}{5} = 7.2\,\Omega$

#19) $\frac{1}{R_T} = \frac{1}{12} + \frac{1}{12} + \frac{1}{12}$ $\frac{1}{R_T} = \frac{3}{12}$ $R_T = \frac{12}{3} = 4\,\Omega$

#20)



a) $\frac{1}{R_T} = \frac{1}{62} + \frac{1}{88} = \frac{88+62}{5412} = \frac{150}{5412}$

$R_T = \frac{5412}{150} = 36\,\Omega$

b) $I_1 = \frac{V}{R} = \frac{12V}{62}$ $I_2 = \frac{V}{R} = \frac{12}{88} = 0.14A$

$I_2 = 0.14A$